

THE DIFFERENTIAL SCATTERING COEFFICIENT FOR DIFFUSE INTERACTIONS

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Abstract. We derive the differential scattering coefficient $\sigma_s(\mathbf{\Omega}' \rightarrow \mathbf{\Omega})$ for a vegetation canopy from the bi-Lambertian model of diffuse leaf scattering. The derivation parallels that of the extinction coefficient and yields the factorization $\sigma_s = u_L \cdot (1/\pi)\Gamma_{LD}$, where u_L is the leaf area density and $(1/\pi)\Gamma_{LD}$ is the area scattering phase function first proposed by Ross (1981). The quantity Γ_{LD} is expressed via a master formula as a linear combination of two fundamental integrals, $\Phi_{| \cdot |}$ and Φ_{full} , whose interplay governs the forward–backward asymmetry of scattering. For uniform leaf inclination and azimuthal distributions, Γ_{LD} reduces to an explicit closed form depending only on the scattering angle. For all other distributions, a single Gauss–Legendre quadrature over the leaf-inclination variable suffices, via the analytical H-function of Shultis and Myneni (1988). The normalization $(1/\pi)\int \Gamma_{LD} d\mathbf{\Omega} = \omega_{LD} G(\mathbf{\Omega}')$ is proved universally, confirming energy conservation. Closed-form results for the six classical leaf inclination distributions under diaheliotropic, paraheliotropic, and uniform azimuthal alignments are collected in the appendices.

§1. Introduction.

In a preceding chapter we derived the *extinction coefficient* $\sigma(\mathbf{r}, \mathbf{\Omega})$, which governs the attenuation of a beam as it traverses the canopy. It satisfies

$$\sigma(\mathbf{r}, \mathbf{\Omega}) = u_L(\mathbf{r}) G(\mathbf{r}, \mathbf{\Omega}).$$

Extinction, however, tells only half the story. A photon that is intercepted by a leaf is not necessarily destroyed: it may be scattered—reflected from the leaf surface or transmitted through the leaf interior—and re-emerge travelling in a new direction. To formulate the complete radiative transfer equation we must therefore specify not only how rapidly photons are removed from a beam, but also how they are redistributed in direction upon each interaction. This is the role of the differential scattering coefficient.

We shall find that the *differential scattering coefficient* admits the factorization

$$\sigma_s(\mathbf{r}, \mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = u_L(\mathbf{r}) \cdot (1/\pi) \Gamma(\mathbf{r}, \mathbf{\Omega}' \rightarrow \mathbf{\Omega}), \quad (1)$$

where u_L is the *leaf area density* introduced earlier and $(1/\pi) \Gamma$ is the *area scattering phase function*, first proposed by Ross (1981). The quantity Γ encapsulates all information about the angular redistribution of scattered radiation by the foliage; it depends upon the leaf-normal orientation distribution, the leaf optical properties (reflectance and transmittance), and the geometry of the

incident and scattered directions. Unlike the extinction coefficient, the differential scattering coefficient depends upon wavelength, because the leaf optical properties are wavelength-dependent.

The derivation proceeds along lines parallel to those of the extinction coefficient. The key new ingredient is the *leaf scattering phase function* γ_L , which describes the angular redistribution of radiation by a single leaf element. In this chapter we first treat the diffuse (bi-Lambertian) component Γ_{LD} arising from scattering within the leaf interior of broadleaf vegetation (Jacquemoud and Baret, 1990).

§2. Definition and derivation of the differential scattering coefficient.

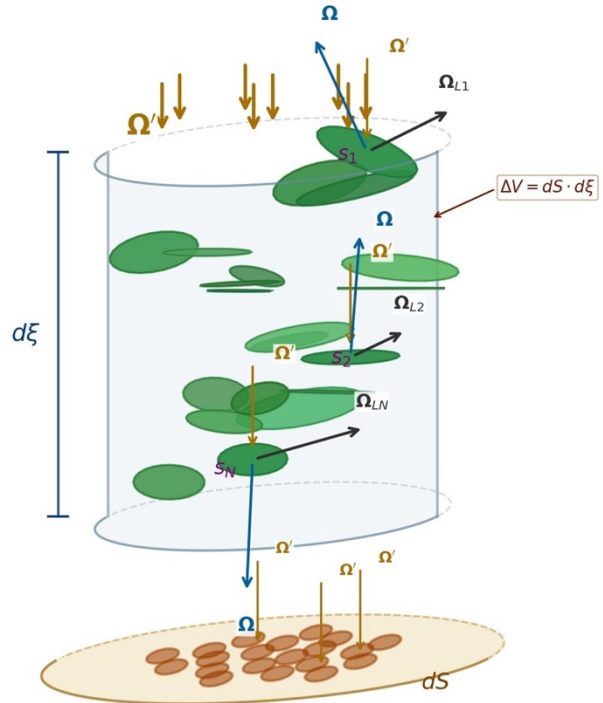
The *differential scattering coefficient* is defined in terms of the probability that a photon, while travelling an infinitesimal distance ξ through the medium, will, upon collision with the host material, scatter from direction Ω' to direction Ω . This probability is given by $\sigma_s(\mathbf{r}, \Omega' \rightarrow \Omega) d\xi d\Omega$. The differential scattering coefficient has dimensions of inverse length per steradian ($\text{m}^{-1} \text{sr}^{-1}$).

Consider, as before, an elementary volume $\Delta V = dS d\xi$ at position $\mathbf{r}$ in the canopy containing N small planar leaf elements. The i th element has one-sided area s_i and outward leaf-normal Ω_{Li} . It intercepts a fraction of the beam proportional to its projected area $s_i |\Omega_{Li} \cdot \Omega'|$ and redistributes the intercepted radiation according to its individual scattering phase function γ_{Li} . Summing over all N elements:

$$\sigma_s(\mathbf{r}, \Omega' \rightarrow \Omega) d\xi d\Omega = (1/dS) \sum_i s_i |\Omega_{Li} \cdot \Omega'| \gamma_{Li}(\Omega' \rightarrow \Omega; \Omega_{Li}) d\Omega, \quad (2)$$

where the interactions are assumed independent and uncorrelated, and the individual shadows do not overlap—the same assumptions invoked for the extinction coefficient.

FIGURE 1. *Geometry of the elementary volume element $\Delta V = dS \cdot d\xi$ used in the derivation of the differential scattering coefficient. The cylindrical volume element ΔV is defined by a cross-sectional area dS oriented perpendicular to the incident photon direction Ω' , and a depth increment $d\xi$ along that direction. Inside the cylinder, N small planar leaf elements are distributed at random positions and orientations. The i th element has one-sided area s_i and outward leaf-normal direction Ω_{Li} . The incident direction Ω' is shown at the top of the cylinder and, locally, at each annotated leaf; the scattered direction Ω is shown emerging from three representative leaves. The projection plane dS receives the shadow cast by each leaf along Ω' ; the shadows are arranged so as not to overlap.*



Counting over individual elements is tedious when N is large. Grouping elements by the direction of their normals and expressing the total area of elements in each solid-angle bin through the *leaf-angle probability density* g_L , the discrete sum becomes a continuous integral as $M \rightarrow \infty$. Identifying $S^0/(dS d\zeta) = u_L(\mathbf{r})$, the *leaf area density*, and defining the *area scattering phase function*

$$\frac{1}{\pi} \Gamma(\mathbf{r}, \boldsymbol{\Omega}' \rightarrow \boldsymbol{\Omega}) = \frac{1}{2\pi} \int_{2\pi} d\boldsymbol{\Omega}_L g_L(\mathbf{r}, \boldsymbol{\Omega}_L) |\boldsymbol{\Omega}_L \cdot \boldsymbol{\Omega}'| \gamma_L(\boldsymbol{\Omega}' \rightarrow \boldsymbol{\Omega}; \boldsymbol{\Omega}_L), \quad (3)$$

we obtain the factorization

$$\sigma_s(\mathbf{r}, \boldsymbol{\Omega}' \rightarrow \boldsymbol{\Omega}) = u_L(\mathbf{r}) \cdot (1/\pi) \Gamma(\mathbf{r}, \boldsymbol{\Omega}' \rightarrow \boldsymbol{\Omega}). \quad (4)$$

In the above, $S^0 = \sum_i s_i$. The result equation (4) parallels the extinction coefficient: u_L describes the *amount* of foliage per unit volume, while $(1/\pi) \Gamma$ describes the *angular pattern* of scattering. Here “leaf area” means one-sided green leaf area in broadleaf vegetation and one-half total needle surface area in coniferous vegetation.

§3. The bi-Lambertian model and the diffuse area scattering phase function.

We adopt the *bi-Lambertian model*, which idealizes diffuse interactions in the interior of a flat leaf as cosine distributions scaled in proportion to the leaf’s *directional-hemispherical reflectance* ρ_{LD} and *transmittance* τ_{LD} (Ross, 1981; Shultis and Myneni, 1988):

$$\gamma_{LD}(\boldsymbol{\Omega}' \rightarrow \boldsymbol{\Omega}; \boldsymbol{\Omega}_L) = \frac{1}{\pi} \rho_{LD} |\boldsymbol{\Omega} \cdot \boldsymbol{\Omega}_L|, \quad (\boldsymbol{\Omega} \cdot \boldsymbol{\Omega}_L)(\boldsymbol{\Omega}' \cdot \boldsymbol{\Omega}_L) < 0, \quad (5a)$$

$$\gamma_{LD}(\boldsymbol{\Omega}' \rightarrow \boldsymbol{\Omega}; \boldsymbol{\Omega}_L) = \frac{1}{\pi} \tau_{LD} |\boldsymbol{\Omega} \cdot \boldsymbol{\Omega}_L|, \quad (\boldsymbol{\Omega} \cdot \boldsymbol{\Omega}_L)(\boldsymbol{\Omega}' \cdot \boldsymbol{\Omega}_L) > 0. \quad (5b)$$

The condition $(\boldsymbol{\Omega} \cdot \boldsymbol{\Omega}_L)(\boldsymbol{\Omega}' \cdot \boldsymbol{\Omega}_L) < 0$ indicates that the incident and scattered photons lie on opposite sides of the leaf (reflection), while the positive condition indicates they lie on the same side (transmission). The *leaf single-scattering albedo* is

$$\omega_{LD} = \rho_{LD} + \tau_{LD}.$$

With this model, the area scattering phase function for diffuse scattering from the leaf interior is (Appendix A, equation (A.3))

$$\Gamma_{LD} = \rho_{LD} \Gamma_{LD}^- + \tau_{LD} \Gamma_{LD}^+, \quad (6)$$

where $\Gamma_{LD}^\pm$ are the reflection (−) and transmission (+) components, defined as integrals over the leaf-normal hemisphere restricted to the domains where the product $(\boldsymbol{\Omega} \cdot \boldsymbol{\Omega}_L)(\boldsymbol{\Omega}' \cdot \boldsymbol{\Omega}_L)$ is negative or positive, respectively. When the leaf inclination θ_L and azimuthal φ_L distributions are independent, the leaf-normal density factorizes as

$$(1/2\pi) g_L(\boldsymbol{\Omega}_L) = (1/2\pi) h_L(\varphi_L) \times \bar{g}_L(\theta_L).$$

§4. The master formula and its properties.

The algebraic structure of Γ_{LD} can be decomposed using two fundamental integrals (Appendix B, equations (B.1)–(B.3)): the *unrestricted integral* Φ^{full} , which can always be evaluated in closed form, and the *absolute-value integral* $\Phi^{|\cdot|}$, which in general requires numerical evaluation. The *master formula* (Appendix B, equation (B.4)) is

$$\Gamma_{LD} = (\omega_{LD}/2) \Phi^{|\cdot|} + ((\tau_{LD} - \rho_{LD})/2) \Phi^{\text{full}}. \quad (7)$$

This decomposition separates Γ_{LD} into a symmetric part (proportional to ω_{LD}) and an antisymmetric part (proportional to $\tau_{\text{LD}} - \rho_{\text{LD}}$) that captures the forward-backward asymmetry induced by unequal reflectance and transmittance. The bi-Lambertian model yields the reciprocity

$$\Gamma_{\text{LD}}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = \Gamma_{\text{LD}}(\mathbf{\Omega} \rightarrow \mathbf{\Omega}').$$

When $\rho_{\text{LD}} = \tau_{\text{LD}}$, the antisymmetric term vanishes and

$$\Gamma_{\text{LD}} = (\omega_{\text{LD}}/2) \Phi^{| \cdot |}.$$

The distinction between reflection and transmission disappears, and an additional symmetry

$$\Gamma_{\text{LD}}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = \Gamma_{\text{LD}}(\mathbf{\Omega}' \rightarrow -\mathbf{\Omega})$$

obtains: a leaf with equal reflectance and transmittance scatters identically into both hemispheres. When $\rho_{\text{LD}} \neq \tau_{\text{LD}}$, the antisymmetric term tilts Γ_{LD} toward forward scattering (if $\tau_{\text{LD}} > \rho_{\text{LD}}$) or backward scattering (if $\rho_{\text{LD}} > \tau_{\text{LD}}$).

For non-uniform $h_{\text{L}}(\varphi_{\text{L}}) = 2 \cos^2(\varphi - \varphi_{\text{L}} - \eta)$, the unrestricted integral acquires an additive correction $\Delta\Phi^{\text{full}}$ (Appendix B, equation (B.6)), while the absolute-value integral requires full numerical evaluation. A notable structural property is the *dia-para symmetry*

$$\Delta\Gamma|_{\text{para}} = -\Delta\Gamma|_{\text{dia}}$$

for every leaf inclination model.

In general, Γ_{LD} is an explicit function of the four angles $(\theta', \varphi', \theta, \varphi)$ and cannot be reduced to a function of the scattering angle β alone (*Non-rotational invariance*). This property precludes the use of Legendre polynomial expansions and the addition theorem typically employed in transport theory.

§5. Normalization and related coefficients.

Integration of the differential scattering coefficient over all scattered directions yields the *total scattering coefficient*

$$\sigma'_{\text{LD}}(\mathbf{r}, \mathbf{\Omega}') = \int_{4\pi} d\mathbf{\Omega} \sigma_{\text{LD}}(\mathbf{r}, \mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = u_{\text{L}}(\mathbf{r}) \omega_{\text{LD}} G(\mathbf{r}, \mathbf{\Omega}'). \quad (8)$$

The absorption coefficient is defined as

$$\sigma_{\text{a}}(\mathbf{r}, \mathbf{\Omega}') = \sigma(\mathbf{r}, \mathbf{\Omega}') - \sigma'_{\text{LD}}(\mathbf{r}, \mathbf{\Omega}') = u_{\text{L}}(\mathbf{r}) G(\mathbf{r}, \mathbf{\Omega}')(1 - \omega_{\text{LD}}),$$

and the single-scattering albedo of the medium for diffuse interactions is

$$\hat{\omega}_{\text{D}} = \sigma'_{\text{LD}}(\mathbf{r}, \mathbf{\Omega}') / \sigma(\mathbf{r}, \mathbf{\Omega}') = \omega_{\text{LD}},$$

provided the leaf albedo does not depend on the leaf-normal direction. A normalized scattering phase function

$$P_{\text{LD}}(\mathbf{r}; \mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = 4 \Gamma_{\text{LD}}(\mathbf{r}, \mathbf{\Omega}' \rightarrow \mathbf{\Omega}) / (\omega_{\text{LD}} G(\mathbf{r}, \mathbf{\Omega}'))$$

satisfies

$$(1/4\pi) \int_{4\pi} d\mathbf{\Omega} P_{\text{LD}}(\mathbf{r}; \mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = 1,$$

for *every* leaf inclination and azimuthal distribution (Appendix B, §10–11). The wavelength dependence cancels in the sum, recovering the wavelength-independent extinction coefficient:

$$\sigma'_{LD}(\mathbf{r}, \boldsymbol{\Omega}') + \sigma_a(\mathbf{r}, \boldsymbol{\Omega}') = u_L(\mathbf{r}) \omega_{LD} G(\mathbf{r}, \boldsymbol{\Omega}') + u_L(\mathbf{r})(1 - \omega_{LD}) G(\mathbf{r}, \boldsymbol{\Omega}') = u_L(\mathbf{r}) G(\mathbf{r}, \boldsymbol{\Omega}') = \sigma(\mathbf{r}, \boldsymbol{\Omega}').$$

This property gives rise to the spectral invariant behaviour of vegetation canopies discussed elsewhere.

§6. The azimuthally-averaged scattering phase function and the H -function.

For the azimuthally-averaged equation in plane-parallel canopies, the relevant quantity is

$$\Gamma_{LD}(\mu' \rightarrow \mu) = (1/2\pi) \int_0^{2\pi} \Gamma_{LD}(\Omega' \rightarrow \Omega) d\varphi \quad (9)$$

where $\mu = \cos \theta$ and $\mu' = \cos \theta'$. The key result of Shultis and Myneni (1988) is that, for uniform h_L , the double azimuthal integration can be performed analytically for any $\bar{g}_L$, reducing Γ_{LD} to a single quadrature:

$$\Gamma_{LD}(\mu' \rightarrow \mu) = \int_0^1 \bar{g}_L(\mu_L) [\tau_{LD} \Psi^+ + \rho_{LD} \Psi^-] d\mu_L \quad (10)$$

where the kernels $\Psi^\pm(\mu, \mu', \mu_L)$ are built from the H -function of Shultis and Myneni (Appendix C, equation (C.11)):

$$\Psi^\pm(\mu, \mu', \mu_L) = H(\mu, \mu_L) H(\pm\mu', \mu_L) + H(-\mu, \mu_L) H(\mp\mu', \mu_L).$$

The H -function is the azimuthally-averaged positive-part projection of a direction onto a leaf normal, a piecewise-analytical expression whose three branches correspond to geometric regimes in which the dot product $\boldsymbol{\Omega} \cdot \boldsymbol{\Omega}_L$ never changes sign, always remains negative, or changes sign at transition angles $\pm\varphi_t$. This reduction is computationally significant: Eq. (10) requires evaluation of the analytical H -function at each of N Gauss–Legendre nodes (typically $N=12$ suffices), compared to the double or triple numerical integration otherwise required. The H -function formulation applies to *uniform* h_L *only*; when the azimuthal leaf-normal distribution is non-uniform, the full Γ_{LD} must be computed numerically via the master formula.

§7. Analytical forms and the moment structure.

Analytical forms of Γ_{LD} depend upon $\bar{g}_L$ and h_L . Complete derivations are given in Appendices C–F; we summarize the principal results here. In the case of $\bar{g}_L = 1$ (uniform) and $h_L = 1$ (uniform) canopies, the area scattering phase function admits the closed form

$$\Gamma_{LD,U} = \omega_{LD}(\sin \beta - \beta \cos \beta)/(3\pi) + \tau_{LD} \cos \beta / 3. \quad (11)$$

This is the only case in which Γ_{LD} depends solely on the scattering angle β and is therefore *rotationally invariant*. For canopies with horizontal leaves ($\theta_0 = 0$), the azimuthal distribution is irrelevant;

$$\Gamma_{LD} = \tau_{LD} |\mu_s \mu'| \quad (\text{transmission}), \quad \Gamma_{LD} = \rho_{LD} |\mu_s \mu'| \quad (\text{reflection}).$$

In the case of canopies with vertical and uniform h_L , a closed form involving the J -function exists (Appendix D, equation (D.15)). For canopies with planophile, erectophile, plagioophile and extremophile distributions of $\bar{g}_L$ and uniform h_L , the H -function formulation of Eq. (10) reduces the problem to a single quadrature.

Full numerical integration of the master formula is required for any $\bar{g}_L$ and non-uniform h_L . The moment structure of the unrestricted integral takes the universal form (Appendix E, §28)

$$\Phi^{\text{full}} = M_2 \mu \mu' + (M_s/2) \sin \theta \sin \theta' \cos \delta + (M_s/4) \sin \theta \sin \theta' \cos(2\eta + \delta), \quad (12)$$

where

$$M_2 = \int_0^\pi \bar{g}_L(\theta_L) \mu_L^2 \sin \theta_L d\theta_L, \quad M_s = 1 - M_2.$$

Values for the classical leaf inclination distributions are given in Table 1.

TABLE 1. Moment structure of the unrestricted integral Φ^{full} for the classical leaf inclination distributions.			
Distribution	$\bar{g}_L$	M_2	M_s
Uniform	$\bar{g}_L = 1$	1/3	2/3
Planophile	$\bar{g}_L = (3/2)(1 + \cos 2\theta_L)$	3/5	2/5
Erectophile	$\bar{g}_L = (3/4)(1 - \cos 2\theta_L)$	1/5	4/5
Plagiophile	$\bar{g}_L = (15/8) \sin^2 2\theta_L$	3/7	4/7
Extremophile	$\bar{g}_L = (15/7) \cos^2 2\theta_L$	11/49	38/49

The full list of analytical expressions for the area scattering phase function for all eight leaf inclination distributions and four azimuthal regimes is given in Table 2.

TABLE 2. Analytical expressions for the area scattering phase function Γ_{LD} for eight leaf inclination distributions under four azimuthal regimes. The Ref. column gives the corresponding appendix letter.					
Distribution	Uniform $h_L = 1$	Diaheliotropic	Paraheliotropic	General η	Ref.*
Horizontal ($\theta_0 = 0$)	$\tau_{LD} \mu_s\mu' $ or $\rho_{LD} \mu_s\mu' $	Same as uniform $h_L^\dagger$	Same as uniform $h_L^\dagger$	Same as uniform $h_L^\dagger$	D
Vertical ($\theta_0 = \pi/2$)	Eq. (D.15)	$\Gamma_U + \Delta\Gamma _{\eta=0}$	$\Gamma_U - \Delta\Gamma _{\eta=0}$	$\Gamma_U + \Delta\Gamma(\eta)$	D
Constant θ_0	$\Psi^\pm$ kernel at θ_0 (single μ_L pt)	$\Gamma_U + \Delta\Gamma _{\eta=0}$	$\Gamma_U - \Delta\Gamma _{\eta=0}$	$\Gamma_U + \Delta\Gamma(\eta)$	D
Uniform	Closed form: $\omega_{LD}(\sin\beta - \beta\cos\beta)/3\pi$ $\tau_{LD}\cos\beta/3$	$\Gamma_U + \Delta\Gamma _{\eta=0}$	$\Gamma_U - \Delta\Gamma _{\eta=0}$	$\Gamma_U + \Delta\Gamma(\eta)$	C
Planophile	$\int_0^1 \bar{g}_L[\tau_{LD}\Psi^+ + \rho_{LD}\Psi^-]d\mu_L \uparrow$	$\Gamma_U + \Delta\Gamma _{\eta=0}$	$\Gamma_U - \Delta\Gamma _{\eta=0}$	$\Gamma_U + \Delta\Gamma(\eta)$	E
Erectophile	$\int_0^1 \bar{g}_L[\tau_{LD}\Psi^+ + \rho_{LD}\Psi^-]d\mu_L \uparrow$	$\Gamma_U + \Delta\Gamma _{\eta=0}$	$\Gamma_U - \Delta\Gamma _{\eta=0}$	$\Gamma_U + \Delta\Gamma(\eta)$	E
Plagiophile	$\int_0^1 \bar{g}_L[\tau_{LD}\Psi^+ + \rho_{LD}\Psi^-]d\mu_L \uparrow$	$\Gamma_U + \Delta\Gamma _{\eta=0}$	$\Gamma_U - \Delta\Gamma _{\eta=0}$	$\Gamma_U + \Delta\Gamma(\eta)$	E
Extremophile	$\int_0^1 \bar{g}_L[\tau_{LD}\Psi^+ + \rho_{LD}\Psi^-]d\mu_L \uparrow$	$\Gamma_U + \Delta\Gamma _{\eta=0}$	$\Gamma_U - \Delta\Gamma _{\eta=0}$	$\Gamma_U + \Delta\Gamma(\eta)$	E

Notes: Universal structure. $\Gamma_{LD} = \Gamma_U + \Delta\Gamma$, where Γ_U is the uniform- h_L result and $\Delta\Gamma$ is the correction from azimuthal anisotropy. **Dia-Para symmetry.** $\Delta\Gamma|_{\text{Para}} = -\Delta\Gamma|_{\text{Dia}}$ for all inclination models. **Horizontal leaves.** Result is $\tau_{LD}|\mu_s\mu'|$ (transmission, same hemisphere) or $\rho_{LD}|\mu_s\mu'|$ (reflection, opposite hemispheres). Azimuthal anisotropy vanishes because $\sin \theta_0 = 0$ eliminates all azimuthal dependence, so all three non-uniform h_L columns equal the

uniform- h_L result. **Non-uniform h_L columns (Dia, Para, General η).** Full numerical integration is required; the H-function is not applicable.

Footnotes

* Ref. column: C = Uniform $\bar{g}_L$; D = Constant inclination; E = Smooth non-uniform (planophile-extremophile); F = Summary of all results.

† H-function formulation of Shultis and Myneni (Appendix C, Eqs. (C.11)–(C.12)). For uniform h_L , the double azimuthal integration is performed analytically, reducing Γ_{LD} to a single quadrature over μ_L with a fully analytical $\Psi_{\pm}$ kernel. For uniform $\bar{g}_L = 1$ this integral yields the closed form in the Uniform row.

‡ For horizontal leaves ($\theta_0 = 0$), $\sin \theta_0 = 0$ eliminates azimuthal dependence; all h_L distributions give the same result.

§8. Illustrative figures.

FIGURE 2. Area scattering phase function Γ_{LD} for uniform leaf inclination ($\bar{g}_L = 1$) and uniform leaf azimuthal distribution ($h_L = 1$), plotted against the cosine of the scattering angle $\cos \beta$. The single-scattering albedo is $\omega_{LD} = 1$, with the transmittance-to-albedo ratio τ_{LD}/ω_{LD} varying from 0 to 0.5. Curves are computed from the analytical closed form equation (C.6). For this combination of uniform $\bar{g}_L$ and uniform h_L , Γ_{LD} depends only on the scattering angle $\cos \beta$ and is therefore rotationally invariant.

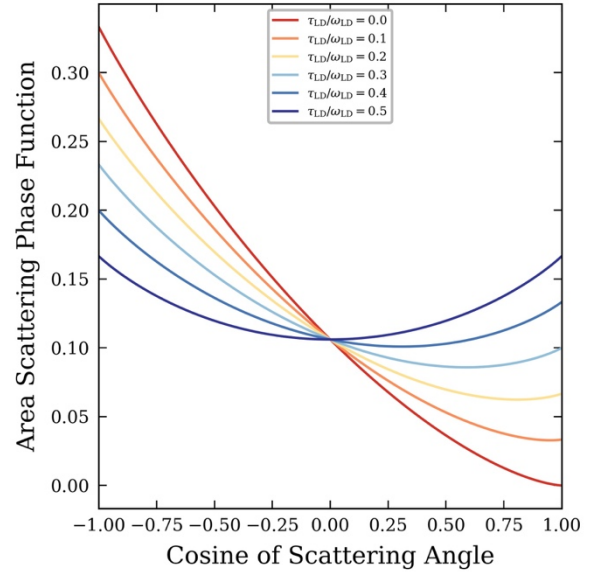


Figure 2 establishes the baseline behaviour of Γ_{LD} under the simplest possible structural assumptions: uniform leaf inclination ($\bar{g}_L = 1$) and uniform leaf azimuthal distribution ($h_L = 1$). In this limiting case the function reduces to the one-dimensional analytical expression equation (C.6) that depends solely on $\cos \beta$, making Γ_{LD} rotationally invariant about the vertical axis. The figure shows the resulting family of smooth curves for unit single-scattering albedo ($\omega_{LD} = 1$) as the transmittance-to-albedo ratio τ_{LD}/ω_{LD} increases from 0 (pure reflection) to 0.5 (equal reflection and transmission). A purely reflective leaf ($\tau_{LD}/\omega_{LD} = 0$) produces strongly asymmetric scattering, with Γ_{LD} rising steeply toward backward scattering ($\cos \beta = -1$) and falling to zero in the forward direction ($\cos \beta = +1$). As the transmittance ratio increases the forward-scattering lobe grows and the backward lobe diminishes; the curves converge to a common crossing point near $\cos \beta \approx 0$, where the reflective and transmissive contributions are balanced regardless of their individual magnitudes. In the equal-contribution limit ($\tau_{LD}/\omega_{LD} = 0.5$) the function is nearly flat and symmetric, reflecting the fact that a leaf which reflects and transmits equally scatters radiation approximately uniformly in all directions. This rotationally invariant, analytically tractable case provides the reference against which the structural sensitivity illustrated in Fig. 3 is assessed.

FIGURE 3. Area scattering phase function Γ_{LD} for the special case $\rho_{LD} = \tau_{LD} = 0.5$ ($\omega_{LD} = 1$), comparing three leaf inclination distributions: uniform (dark dots; the analytical closed form is given by Eq. (C.6) and shown as the solid black curve), planophile (red dots), and erectophile (blue dots), all computed with a uniform leaf azimuthal distribution ($h_L = 1$) and incident direction $\theta' = 135^\circ$, $\phi' = 0^\circ$. Each dot represents Γ_{LD} evaluated at a Gauss-quadrature node (θ_s, ϕ_s) and plotted against the cosine of the scattering angle $\cos \beta$.

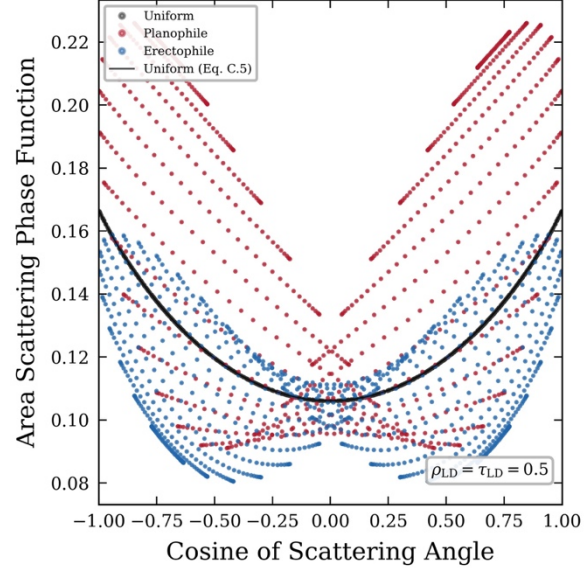


Figure 3 probes the breakdown of rotational invariance when the leaf inclination distribution departs from uniformity. Retaining the equal-contribution leaf optics ($\rho_{LD} = \tau_{LD} = 0.5$, $\omega_{LD} = 1$) and the uniform leaf azimuthal distribution ($h_L = 1$), the function Γ_{LD} is evaluated numerically at each Gauss-quadrature node (θ_s, ϕ_s) in the scattered hemisphere and plotted against $\cos \beta$ for three leaf inclination distributions: uniform, planophile (leaves preferentially horizontal), and erectophile (leaves preferentially vertical). For the uniform distribution the numerical points coincide precisely with the analytical curve of Eq. (C.6), confirming that the quadrature correctly resolves the integral. For the planophile and erectophile distributions, however, the single-valued relationship between Γ_{LD} and $\cos \beta$ disintegrates: at each value of $\cos \beta$ the points spread into a fan of distinct strands, each corresponding to a fixed polar angle θ_s with varying azimuth ϕ_s . This fanning is not a numerical artefact but an intrinsic geometric consequence of the inclination weighting. Once the leaf-normal distribution concentrates normals away from the isotropic, the contribution of a leaf element to Γ_{LD} depends separately on θ_s and ϕ_s , not only on the angle between the incident and scattered directions. The planophile distribution shifts the strands to higher values of Γ_{LD} relative to the erectophile distribution, because near-horizontal leaves intercept predominantly downwelling radiation more efficiently. The figure therefore makes visible a key modelling implication: for realistic vegetation canopies, where planophile or erectophile inclination distributions are common, a parameterisation of Γ_{LD} based solely on the scattering angle β is inadequate, and the full four-angle dependence ($\theta', \phi', \theta_s, \phi_s$) must be retained.

Taken together, Figs. 2 and 3 delineate the conditions under which Γ_{LD} can be treated as a function of the scattering angle alone—and hence represented compactly by a one-dimensional curve—versus those under which the full directional dependence is indispensable. The transition is governed entirely by the leaf inclination distribution: the uniform case is the unique configuration in which the azimuthal integration over all leaf orientations yields a result that is independent of the scattered azimuth ϕ_s . Any departure from uniformity, whether toward planophile or erectophile geometry, breaks this symmetry and introduces a dependence on the full four-angle geometry. From a practical standpoint this distinction has direct consequences for radiative-transfer model complexity: rotationally invariant phase functions admit analytical

quadrature and efficient tabulation, whereas non-invariant distributions require either high-order numerical quadrature in the full four-dimensional solid-angle space or approximate decoupling schemes. Figures 2 and 3 provide a clear visual motivation for that additional complexity.

FIGURE 4. Polar plots of the weighted diffuse area scattering phase function Γ_{LD} . Columns: uniform $\bar{g}_L$ (left) and planophile $\bar{g}_L$ (right), both with non-uniform h_L ($\eta = 55^\circ$). Rows: weighted reflection component $\rho_{LD} \Gamma_{LD}^-$ (top); weighted transmission component $\tau_{LD} \Gamma_{LD}^+$ (bottom). These components are defined in Appendix A. Leaf optical properties: $\rho_{LD} = 0.55$, $\tau_{LD} = 0.45$ ($\omega_{LD} = 1.00$). The polar coordinate is the scattered zenith angle θ_s ; the azimuthal coordinate is the scattered azimuth φ_s , increasing clockwise from north (0°). Incident direction: $\theta' = 140^\circ$, $\varphi' = 0^\circ$. The white star marks retro-incident direction ($\theta_s = 40^\circ$, $\varphi_s = 180^\circ$).

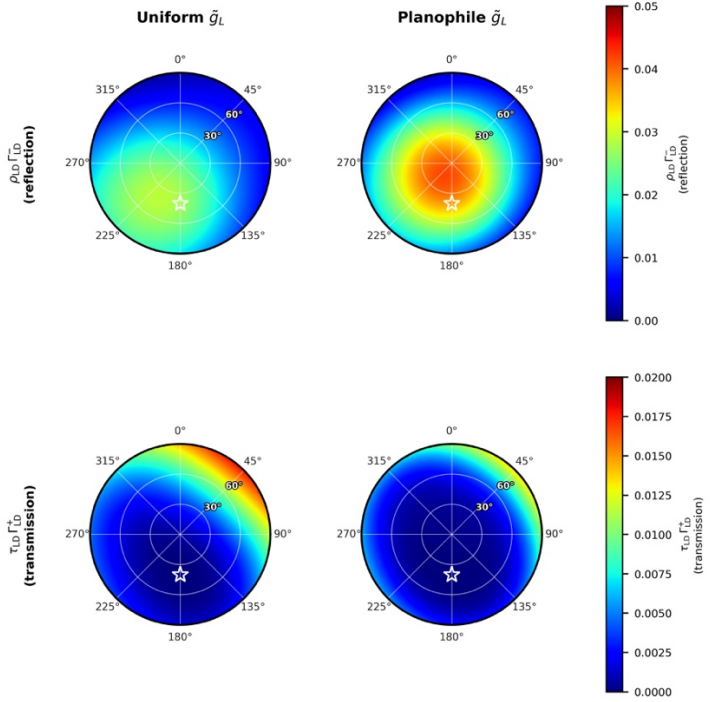


Figure 4 presents polar plots of the two weighted components of the diffuse area scattering phase function, $\rho_{LD} \Gamma_{LD}^-$ and $\tau_{LD} \Gamma_{LD}^+$ which arise from the sign condition on the dot-product product $(\mathbf{\Omega} \cdot \mathbf{\Omega}_L)(\mathbf{\Omega}' \cdot \mathbf{\Omega}_L)$. When this product is negative the incident and scattered rays lie on the same sides of the leaf, and the interaction is a reflection weighted by ρ_{LD} ; when it is positive they lie on the opposite sides and the interaction is a transmission weighted by τ_{LD} . Because $\rho_{LD} > \tau_{LD}$ in the present example, the reflection panels carry higher absolute values throughout the hemisphere. For any leaf inclination distribution the two components are related by an exact azimuthal complementarity: the domain S^- that generates reflection and the domain S^+ that generates transmission are mapped into each other by reversing the sign of $\mathbf{\Omega}$, so every local maximum of Γ_{LD}^- at azimuth $\varphi_s = \alpha$ corresponds to a local maximum of Γ_{LD}^+ at $\varphi_s = \alpha + 180^\circ$. This 180° azimuthal shift is a rigorous consequence of the bi-Lambertian phase function and is visible by direct comparison of the top and bottom rows in the figure.

In the reflection row the scattered energy is concentrated in the azimuthal sector centred near $\varphi_s \approx 210^\circ - 215^\circ$. This can be understood as follows. The preferred leaf-normal azimuth of h_L has two lobes, at $\varphi_L \approx 55^\circ$ and $\varphi_L \approx 235^\circ$. The magnitude of the incident projection $|\mathbf{\Omega}' \cdot \mathbf{\Omega}_L|$ is larger for the 235° lobe than for the 55° lobe, because the incident direction ($\varphi' = 0^\circ$, roughly northward) subtends a larger angle with a leaf normal pointing south-southwest than with one pointing north-northeast. This lobe therefore carries the larger cross-section weight in the integral defining Γ_{LD}^- , and it preferentially scatters energy back toward azimuths close to 235° . The planophile column amplifies this substantially: concentrating $\bar{g}_L$ near $\mu_L \approx 1$ (horizontal leaves) increases the projected

leaf area $|\mathbf{\Omega}' \cdot \mathbf{\Omega}_L|$ for the near-grazing incident ray (surface angle 40°), so the same angular pattern is reproduced with roughly twice the amplitude and a steeper radial gradient. The radial maximum of $\rho_{LD} \Gamma_{LD}$ lies at moderate polar angles ($\theta_s \approx 30^\circ - 50^\circ$) where both $|\mathbf{\Omega} \cdot \mathbf{\Omega}_L|$ and $|\mathbf{\Omega}' \cdot \mathbf{\Omega}_L|$ are simultaneously large for near-horizontal leaf normals; the planophile distribution weights precisely these normals most heavily.

The transmission row also shows the complementary pattern. The dominant leaf-normal lobe at $\varphi_L \approx 235^\circ$ now requires the scattered direction to be on the opposite side of the leaf as the incident ray, i.e. $\mathbf{\Omega}$ must have a dot product with $\mathbf{\Omega}_L$ of the same sign as $\mathbf{\Omega}' \cdot \mathbf{\Omega}_L$. For a downward incident ray ($\mathbf{\Omega}' \cdot \mathbf{\Omega}_L < 0$ for upper-hemisphere leaf normals) the transmitted ray must also have $\mathbf{\Omega} \cdot \mathbf{\Omega}_L < 0$, which places $\mathbf{\Omega}$ in the lower hemisphere. The azimuth of $\mathbf{\Omega}$ that maximises $|\mathbf{\Omega} \cdot \mathbf{\Omega}_L|$ for a leaf normal at $\varphi_L \approx 235^\circ$ is roughly opposite in azimuth to that normal, i.e. $\varphi_s \approx 55^\circ$, rotated toward the preferred orientation $\eta = 55^\circ$ from the pure forward-scatter direction $\varphi_s = 0^\circ$. The transmission peak therefore lands near $\varphi_s \approx 40^\circ - 50^\circ$, on the same side of the disc as the incident direction but displaced clockwise by η . It is a purely geometric consequence of the bi-Lambertian phase function, in which the amplitude in any scattered direction $\mathbf{\Omega}$ is proportional to $|\mathbf{\Omega} \cdot \mathbf{\Omega}_L|$. As with reflection, the planophile distribution raises the peak values while narrowing the radial spread relative to the uniform case.

The preferred azimuth $\eta = 55^\circ$ thus acts as a pivot that rotates both the reflection and transmission peaks simultaneously away from the pure backscatter–forward-scatter axis defined by $\varphi' = 0^\circ$. Were h_L uniform, rotational symmetry about the vertical axis would hold and Γ_{LD} would depend only on θ_s and the scattering angle; the azimuthal patterns seen in Fig. 4 would collapse to rotationally symmetric discs. The combined effect of inclination anisotropy ($\bar{g}_L$) and azimuthal anisotropy (h_L) is therefore to introduce a three-way coupling among the incident azimuth φ' , the preferred leaf-normal azimuth η , and the scattered azimuth φ_s that cannot be captured by any one-dimensional approximation to Γ_{LD} . In radiative-transfer codes that approximate Γ_{LD} by its azimuthal average, the angular-redistribution error introduced by this three-way coupling is largest for canopies with a strong preferred leaf orientation, precisely the condition prevailing in row crops, wind-aligned forests, and many grass and sedge species.

§9. Concluding remarks.

This chapter has developed the differential scattering coefficient $\sigma_s(\mathbf{\Omega}' \rightarrow \mathbf{\Omega})$ for a vegetation canopy from first principles, beginning with the bi-Lambertian model of diffuse leaf scattering and arriving at the compact factorization $\sigma_s = u_L \cdot (1/\pi) \Gamma_{LD}$. The central analytic result is a master formula (Appendix B) that decomposes Γ_{LD} into two fundamental integrals, $\Phi|\cdot|$ and Φ^{full} , whose relative weight is controlled by the leaf single-scattering albedo ω_{LD} and the reflectance–transmittance asymmetry $\tau_{LD} - \rho_{LD}$. For a uniform leaf inclination distribution (Appendix C), both integrals reduce to closed form and Γ_{LD} depends only on the scattering angle β . For all other distributions—constant, planophile, erectophile, plagiophile, and extremophile (Appendices D and E)—the H-function of Shultis and Myneni (1988) reduces the remaining θ_L integration to a single Gauss–Legendre quadrature, rendering the full four-angle phase function computationally tractable. A universal normalization identity confirms that energy conservation is satisfied for every properly normalized inclination model.

The illustrative figures of §8 reveal two qualitatively distinct regimes. When the leaf inclination distribution is uniform, Γ_{LD} is rotationally invariant and admits a compact one-

dimensional representation in $\cos \beta$. Any departure from uniformity—whether planophile, erectophile, or otherwise—introduces a three-way coupling among incident azimuth, preferred leaf-normal azimuth, and scattered azimuth that cannot be captured by any scattering-angle parameterisation. For canopies with a strong preferred leaf orientation, such as row crops, wind-aligned forests, and many grass species, the full four-angle dependence of Γ_{LD} must be retained in any model that aims to resolve azimuthal anisotropy in the radiation field. The analytical framework developed here, together with the closed-form and H-function results collected in Appendix F, provides the complete set of building blocks needed for such computations.

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APPENDIX DERIVATIONS OF THE AREA SCATTERING PHASE FUNCTION

Ranga B. Myneni



CONTENTS

Appendix A. Definitions, Notation, and Conventions · Appendix B. The Master Formula and Universal Normalization · Appendix C. Uniform Leaf Inclination · Appendix D. Constant Leaf Inclination · Appendix E. Smooth Non-Uniform Leaf Inclination Distributions · Appendix F. Summary of All Results

Abstract. Six appendices provide the complete analytical derivations supporting the results of this chapter. Appendix A establishes the definitions, notation, and bi-Lambertian phase function conventions used throughout. Appendix B derives the master formula decomposing Γ_{LD} into the fundamental integrals $\Phi_{| \cdot |}$ and Φ_{full} and proves the universal normalization identity via Fubini's theorem. Appendix C derives the closed-form expression for Γ_{LD} under uniform leaf inclination, the H-function representation, and the dia-para symmetry properties. Appendix D treats the constant (delta-function) inclination case, obtaining exact results for arbitrary and horizontal-vertical leaf orientations. Appendix E gives the analytical forms of Γ_{LD} for the four smooth non-uniform distributions—planophile, erectophile, plagiophile, and extremophile—expressed through the moment parameters M_2 and M_s . Appendix F collects all principal results into a single reference table.

APPENDIX A. DEFINITIONS, NOTATION, AND CONVENTIONS

§1. The Area Scattering Phase Function

$$\frac{1}{\pi} \Gamma(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = \frac{1}{2\pi} \int_{2\pi} d\mathbf{\Omega}_L g_L(\mathbf{\Omega}_L) |\mathbf{\Omega}_L \cdot \mathbf{\Omega}'| \gamma_L(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}; \mathbf{\Omega}_L), \quad (A.1)$$

where $\mathbf{\Omega}'$ is the incident direction, $\mathbf{\Omega}$ the scattered direction, $\mathbf{\Omega}_L$ the outward leaf normal, and g_L the *leaf-normal orientation density*. The *bi-Lambertian leaf phase function* is:

$$\gamma_{LD}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}; \mathbf{\Omega}_L) = \begin{cases} \pi^{-1} \rho_{LD} |\mathbf{\Omega} \cdot \mathbf{\Omega}_L|, & (\mathbf{\Omega} \cdot \mathbf{\Omega}_L)(\mathbf{\Omega}' \cdot \mathbf{\Omega}_L) < 0 \\ \pi^{-1} \tau_{LD} |\mathbf{\Omega} \cdot \mathbf{\Omega}_L|, & (\mathbf{\Omega} \cdot \mathbf{\Omega}_L)(\mathbf{\Omega}' \cdot \mathbf{\Omega}_L) > 0 \end{cases} \quad (A.2)$$

Here ρ_{LD} is the leaf reflectance, τ_{LD} the leaf transmittance, and $\omega_{LD} = \rho_{LD} + \tau_{LD}$ the *single-scattering albedo*.

§2. Structure of Γ_{LD}

$$\Gamma_{LD}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = \rho_{LD} \Gamma_{LD}^-(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) + \tau_{LD} \Gamma_{LD}^+(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) \quad (A.3)$$

$$\Gamma_{LD}^\pm(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = \pm \frac{1}{2\pi} \int_0^{\pi/2} \sin \theta_L d\theta_L \int_{S^\pm} d\varphi_L g_L(\mathbf{\Omega}_L) (\mathbf{\Omega} \cdot \mathbf{\Omega}_L)(\mathbf{\Omega}' \cdot \mathbf{\Omega}_L). \quad (A.4)$$

§3. Notation

The leaf-normal distribution factorizes as $g_L = \bar{g}_L(\theta_L) \cdot h_L(\varphi_L)$ with

$$\int_0^{\pi/2} \bar{g}_L(\theta_L) \sin\theta_L d\theta_L = 1, \quad \frac{1}{2\pi} \int_0^{2\pi} h_L(\varphi_L) d\varphi_L = 1 \quad . \quad (\text{A.5})$$

We write $\mu = \cos\theta$, $\mu' = \cos\theta'$, $\mu_L = \cos\theta_L$. The non-uniform azimuthal distribution is $h_L = 2\cos^2(\varphi - \varphi_L - \eta)$. The dot products are:

$$\mathbf{\Omega} \cdot \mathbf{\Omega}_L = a + b\cos\alpha, \quad \mathbf{\Omega}' \cdot \mathbf{\Omega}_L = a' + b'\cos(\alpha - \delta) \quad (\text{A.6})$$

with $\alpha = \varphi_L - \varphi$, $\delta = \varphi' - \varphi$, and

$$a = \mu\mu_L, \quad b = \sin\theta\sin\theta_L, \quad a' = \mu'\mu_L, \quad b' = \sin\theta'\sin\theta_L. \quad (\text{A.7})$$

§4. Scattering Angle

$$\cos\beta = \mu\mu' + \sin\theta\sin\theta'\cos\delta \quad . \quad (\text{A.8})$$

§5. Normalization

$$\frac{1}{\pi} \int_{4\pi} \Gamma_{\text{LD}}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) d\mathbf{\Omega} = \omega_{\text{LD}} G(\mathbf{\Omega}'). \quad (\text{A.9})$$

APPENDIX B. THE MASTER FORMULA AND UNIVERSAL NORMALIZATION

The key integrals Φ^{full} and $\Phi^{|\cdot|}$ arise from splitting the sign conditions on the dot products $\mathbf{\Omega} \cdot \mathbf{\Omega}_L$ and $\mathbf{\Omega}' \cdot \mathbf{\Omega}_L$.

§1. Key Integrals

$$\Phi^{\text{full}} = \frac{1}{2\pi} \int_0^{2\pi} h_L(\alpha) (a + b\cos\alpha)(a' + b'\cos(\alpha - \delta)) d\alpha \quad (\text{B.1})$$

$$\Phi^{|\cdot|} = \frac{1}{2\pi} \int_0^{2\pi} h_L(\alpha) |a + b\cos\alpha| |a' + b'\cos(\alpha - \delta)| d\alpha \quad . \quad (\text{B.2})$$

Solving the sign decomposition:

$$\Gamma_{\text{ker}}^+(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = \frac{1}{2}(\Phi^{|\cdot|} + \Phi^{\text{full}}), \quad \Gamma_{\text{ker}}^-(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = \frac{1}{2}(\Phi^{|\cdot|} - \Phi^{\text{full}}) \quad (\text{B.3})$$

Master formula:

$$\Gamma_{\text{LD,ker}}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = \frac{\omega_{\text{LD}}}{2} \Phi^{|\cdot|} + \frac{\tau_{\text{LD}} - \rho_{\text{LD}}}{2} \Phi^{\text{full}} \quad (\text{B.4})$$

§2. Φ^{full} for Uniform h_L

$$\Phi_{\text{U}}^{\text{full}} = \mu\mu'\mu_L^2 + \frac{1}{2}\sin\theta\sin\theta'\sin^2\theta_L\cos\delta \quad . \quad (\text{B.5})$$

§3. Non-Uniform Correction

$$\Delta\Phi^{\text{full}} = \frac{1}{4} \sin\theta \sin\theta' \sin^2\theta_L \cos(2\eta + \delta) \quad (\text{B.6})$$

$$\Phi_{\text{NU}}^{\text{full}} = \mu\mu'\mu_L^2 + \frac{1}{2} \sin\theta \sin\theta' \sin^2\theta_L \cos\delta + \frac{1}{4} \sin\theta \sin\theta' \sin^2\theta_L \cos(2\eta + \delta). \quad (\text{B.7})$$

§4. Special Case: $\rho_{\text{LD}} = \tau_{\text{LD}}$

$$\omega_{\text{LD}} = 2\rho_{\text{LD}} \text{ and } \tau_{\text{LD}} - \rho_{\text{LD}} = 0,$$

$$\Gamma_{\text{LD}}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = \frac{\omega_{\text{LD}}}{2} \Phi^{|\cdot|}, \quad (\text{B.8})$$

The unrestricted integral Φ^{full} drops out entirely. This is a significant simplification: the distinction between reflection and transmission disappears, and Γ_{LD} depends only on the absolute-value scattering integral. Since $\Phi^{|\cdot|}$ is inherently non-negative, Eq. (B.8) makes manifest that $\Gamma_{\text{LD}} \geq 0$. For each leaf inclination model, Eq. (B.8) gives:

Uniform $\bar{g}_L$, uniform h_L :

$$\Gamma_{\text{LD}}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = \frac{\omega_{\text{LD}}}{2} \left[\frac{2}{3\pi} (\sin\beta - \beta\cos\beta) + \frac{\cos\beta}{3} \right] \quad (\text{B.9})$$

Constant $\bar{g}_L$, horizontal leaves ($\theta_0 = 0$):

$$\Gamma_{\text{LD}}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = \frac{\omega_{\text{LD}}}{2} |\mu\mu'| \quad (\text{B.10})$$

This unifies the two branches of Eq. (D.11) into a single expression.

Constant $\bar{g}_L$, vertical leaves ($\theta_0 = \pi/2$), uniform h_L :

$$\Gamma_{\text{LD}}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = \frac{\omega_{\text{LD}}}{2} \sin\theta \sin\theta' J(\delta) \quad (\text{B.11})$$

General $\bar{g}_L$, non-uniform h_L :

$$\Gamma_{\text{LD}}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = \frac{\omega_{\text{LD}}}{2} \int_0^{\pi/2} \bar{g}_L(\theta_L) \phi_{\text{NU}} \sin\theta_L d\theta_L \quad (\text{B.12})$$

Physical interpretation. A leaf with $\rho_{\text{LD}} = \tau_{\text{LD}}$ scatters equally into both hemispheres regardless of which side the photon strikes. The scattered radiance depends only on the total interception $|\mathbf{\Omega} \cdot \mathbf{\Omega}_L| |\mathbf{\Omega}' \cdot \mathbf{\Omega}_L|$, not on the relative orientation of incident and scattered beams with respect to the leaf surface.

§5. Theorem. For any $\bar{g}_L, \mathbf{h}_L$ satisfying Eq. (A.5): Eq. (A.9) holds.

§6. Proof. On the same-sign hemisphere, $(\mathbf{\Omega} \cdot \mathbf{\Omega}_L)(\mathbf{\Omega}' \cdot \mathbf{\Omega}_L) = |\mathbf{\Omega} \cdot \mathbf{\Omega}_L| |\mathbf{\Omega}' \cdot \mathbf{\Omega}_L|$. Integrating over $\mathbf{\Omega}$ (Fubini), $|\mathbf{\Omega}' \cdot \mathbf{\Omega}_L|$ factors out. The hemisphere integral gives $\int_{\text{hemi}} |\mathbf{\Omega} \cdot \mathbf{\Omega}_L| d\mathbf{\Omega} = \pi$, so each hemisphere contributes 1. Therefore,

$$\frac{1}{\pi} \int_{4\pi} \Gamma_{\text{LD}}^{\pm}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) d\mathbf{\Omega} = G(\mathbf{\Omega}'), \frac{1}{\pi} \int_{4\pi} \Gamma_{\text{LD}}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) d\mathbf{\Omega} = \omega_{\text{LD}} G(\mathbf{\Omega}'). \quad \checkmark$$

APPENDIX C. UNIFORM LEAF INCLINATION

For uniform leaf inclination the absolute-value integral $\Phi|\cdot|$ can be evaluated analytically by aligning the polar axis with the incident direction Ω' .

§1. $\bar{g}_L(\theta_L) = 1$; normalization: $\int_0^{\pi/2} \sin \theta_L d\theta_L = 1$. ✓

§2. Unrestricted Integral after θ_L Integration

Using $\int_0^{\pi/2} \mu_L^2 \sin \theta_L d\theta_L = \frac{1}{3}$ and $\int_0^{\pi/2} \sin^2 \theta_L \sin \theta_L d\theta_L = \frac{2}{3}$:

$$\Gamma_{LD}^{\text{full}} = \frac{\mu\mu'}{3} + \frac{\sin\theta\sin\theta'}{3} \cos\delta + \frac{\sin\theta\sin\theta'}{6} \cos(2\eta + \delta) \quad (\text{C.1})$$

$$\Gamma_{LD}^{\text{full}} = \frac{1}{3} \cos\beta \quad (\text{C.2})$$

$$\Delta\Gamma_{LD}^{\text{full}} = \frac{\sin\theta\sin\theta'}{6} \cos(2\eta + \delta). \quad (\text{C.3})$$

§3. The Absolute-Value Integral $\Phi_U^{|\cdot|}$

Aligning the polar axis with Ω' so that $|\Omega' \cdot \Omega_L| = \mu'_L$ and the azimuthal integral reduces to $\psi_U(\cos\beta, \mu'_L)$ (G-function Appendix, Eqs. (B.6a,b)):

$$\Phi_U^{|\cdot|} = \int_0^{\pi/2} \psi_U(\cos\beta, \mu'_L) \mu'_L \sin\theta'_L d\theta'_L \quad (\text{C.4})$$

Branch point at $\theta_L'^* = \pi/2 - \beta$. Evaluating by regions:

$$I_1 = \frac{\cos\beta}{3} (1 - \sin^3\beta), \quad I_2 = \frac{2}{3\pi} (\sin\beta - \beta\cos\beta) - \frac{\cos\beta}{3} (1 - \sin^3\beta) + \frac{\cos\beta}{3} \sin^3\beta$$

Combining:

$$\Phi_U^{|\cdot|} = \frac{2}{3\pi} (\sin\beta - \beta\cos\beta) + \frac{\cos\beta}{3} \quad (\text{C.5})$$

Check: $\Phi_U^{|\cdot|}(0) = \frac{1}{3}$, $\Phi_U^{|\cdot|}(\pi/2) = \frac{2}{3\pi}$, $\Phi_U^{|\cdot|}(\pi) = \frac{1}{3}$. ✓

§4. Closed Form

$$\omega_{LD} + \tau_{LD} - \rho_{LD} = 2\tau_{LD}$$

$$\Gamma_{LD,U} = \frac{\omega_{LD}}{3\pi} (\sin\beta - \beta\cos\beta) + \frac{\tau_{LD}}{3} \cos\beta \quad . \quad (\text{C.6})$$

§5. Non-Uniform h_L

$$\Gamma_{LD} = \frac{\omega_{LD}}{2} \Phi^{|\cdot|} + \frac{\tau_{LD} - \rho_{LD}}{2} \Gamma_{LD}^{\text{full}}, \quad (\text{C.7})$$

with $\Phi^{||} = \int_0^{\pi/2} \phi_{NU} \sin\theta_L d\theta_L$ and $\phi_{NU} = \frac{1}{2\pi} \int_0^{2\pi} h_L |a + b\cos\alpha| |a' + b'\cos(\alpha - \delta)| d\alpha$.

§6. Direct Normalization

Using $d\Omega = \sin\beta d\beta d\varphi_\beta$ and $\int_0^\pi \sin\beta(\sin\beta - \beta\cos\beta) d\beta = \frac{3\pi}{4}$:

$$\frac{1}{\pi} \int_{4\pi} \Gamma_{LD,U} d\Omega = 2 \cdot \frac{\omega_{LD}}{3\pi} \cdot \frac{3\pi}{4} = \frac{\omega_{LD}}{2} = \omega_{LD} G_U \quad \checkmark$$

§7. Uniform $\mathbf{h}_L$: $\Gamma_{LD,U} = \frac{\omega_{LD}}{3\pi}(\sin\beta - \beta\cos\beta) + \frac{\tau_{LD}}{3}\cos\beta$

§8. Diaheliotropic ($\boldsymbol{\eta} = \mathbf{0}$):

$$\Gamma_{LD}^{\text{full}}|_{\eta=0} = \frac{\mu\mu'}{3} + \frac{\sin\theta\sin\theta'}{2}\cos\delta = \frac{1}{3}\cos\beta + \frac{\sin\theta\sin\theta'}{6}\cos\delta$$

§9. Paraheliotropic ($\boldsymbol{\eta} = \boldsymbol{\pi}/2$):

$$\Gamma_{LD}^{\text{full}}|_{\eta=\pi/2} = \frac{\mu\mu'}{3} + \frac{\sin\theta\sin\theta'}{6}\cos\delta = \frac{1}{3}\cos\beta - \frac{\sin\theta\sin\theta'}{6}\cos\delta$$

Dia–Para symmetry: $\Delta\Gamma_{LD}^{\text{full}}|_{\eta=\pi/2} = -\Delta\Gamma_{LD}^{\text{full}}|_{\eta=0}$. $\checkmark$

§10. Motivation

The complete area scattering phase function $\Gamma_{LD}(\Omega' \rightarrow \Omega_s)$ depends on four angles $(\theta', \varphi', \theta_s, \varphi_s)$. For uniform h_L , only the relative azimuth $\varphi = \varphi_s - \varphi'$ matters (three independent variables). In many radiative transfer formulations — particularly the azimuthally-averaged equation for plane-parallel canopies — the quantity of interest is the azimuthally-averaged scattering phase function:

$$\Gamma_{LD}(\mu' \rightarrow \mu) = \frac{1}{2\pi} \int_0^{2\pi} d\varphi \Gamma_{LD}(\boldsymbol{\Omega}' \rightarrow \boldsymbol{\Omega}_s) \quad (\text{C.8})$$

where $\mu = \cos\theta$ and $\mu' = \cos\theta'$. The subscript d denotes dependence on the leaf-normal distribution.

The key result of Shultis and Myneni (1988) is that the double azimuthal integration over (φ, φ_L) can be performed analytically for any leaf inclination distribution $\bar{g}_L$, reducing Γ_d to a single integral over $\mu_L = \cos\theta_L$.

§11. The Ψ -Kernel

The azimuthally-averaged scattering phase function can be written as:

$$\Gamma_{LD}(\mu' \rightarrow \mu) = \int_0^1 d\mu_L \bar{g}_L(\mu_L) [\tau_{LD} \Psi^+(\mu, \mu', \mu_L) + \rho_{LD} \Psi^-(\mu, \mu', \mu_L)] \quad (C.9)$$

where $\Psi^\pm$ are the azimuthally-averaged scattering kernels:

$$\Psi^\pm(\mu, \mu', \mu_L) = (\pm) \frac{1}{4\pi^2} \int_0^{2\pi} d\varphi \int_0^{2\pi} d\varphi_L (\boldsymbol{\Omega}' \cdot \boldsymbol{\Omega}_L)(\boldsymbol{\Omega} \cdot \boldsymbol{\Omega}_L) \quad (C.10)$$

with the $(\pm)$ sign enforcing the sign convention: Ψ^+ corresponds to the region where $(\boldsymbol{\Omega}' \cdot \boldsymbol{\Omega}_L)$ and $(\boldsymbol{\Omega} \cdot \boldsymbol{\Omega}_L)$ have the same sign (transmission), and Ψ^- where they have opposite signs (reflection).

§12. Analytical Evaluation: The H-Function

The double azimuthal integral in Eq. (C.10) can be evaluated analytically (Shultis and Myneni, 1988). The result is:

$$\Psi^\pm(\mu, \mu', \mu_L) = H(\mu, \mu_L) H(\pm\mu', \mu_L) + H(-\mu, \mu_L) H(\mp\mu', \mu_L) \quad (C.11)$$

where the H-function is the piecewise expression:

$$H(\mu, \mu_L) = \begin{cases} \mu \mu_L, & \cot\theta \cot\theta_L > 1, \\ 0, & \cot\theta \cot\theta_L < -1, \\ \frac{1}{\pi} \left[\mu \mu_L \varphi_t(\mu) + \sqrt{1 - \mu^2} \sqrt{1 - \mu_L^2} \sin\varphi_t(\mu) \right], & \text{otherwise} \end{cases} \quad (C.12)$$

with the transition angle:

$$\sin\varphi_t(\mu) = -\cot\theta \cot\theta_L, \quad \varphi_t(\mu) = \pi - \varphi_t(-\mu). \quad (C.13)$$

Remark. The three branches of the H-function correspond to geometric configurations. Branch 1 ($\cot\theta \cdot \cot\theta_L > 1$): the beam and leaf normal are both near-vertical on the same side; the dot product $\boldsymbol{\Omega} \cdot \boldsymbol{\Omega}_L$ does not change sign over $0 \leq \varphi_L \leq 2\pi$. Branch 2 ($\cot\theta \cdot \cot\theta_L < -1$): beam and leaf normal are on opposite sides and the dot product is always negative; the positive projection vanishes. Branch 3 (otherwise): the dot product changes sign at the transition angles $\pm\varphi_t$, and the integral splits into positive and negative contributions.

Note that the H-function is precisely related to the G-function's uniform azimuthal projection integral ψ_U (Eqs. B.6a,b of the G-function Appendix): $H(\mu, \mu_L) = \psi_U(\mu_L, \mu) / 2\pi$ evaluated at a single leaf inclination.

§13. Reduction to a Single Integral

Combining Eqs. (C.9) and (C.11), the azimuthally-averaged scattering phase function for any leaf inclination distribution $\bar{g}_L$ with uniform h_L is:

$$\Gamma_{LD}(\mu' \rightarrow \mu) = \int_0^1 \bar{g}_L(\mu_L) [\tau_{LD} (H(\mu, \mu_L) H(\mu', \mu_L) + H(-\mu, \mu_L) H(-\mu', \mu_L)) + \rho_{LD} (H(\mu, \mu_L) H(-\mu', \mu_L) + H(-\mu, \mu_L) H(\mu', \mu_L))] d\mu_L \quad (C.14)$$

This is the central result: for uniform h_L , Γ_{LD} is reduced from a triple integral to a single quadrature over μ_L with a fully analytical kernel.

§14. Recovery of the Uniform Closed Form

For uniform $\bar{g}_L = 1$, Eq. (C.14) becomes:

$$\Gamma_{LD}(\mu' \rightarrow \mu) = \int_0^1 [\tau_{LD} \Psi^+ + \rho_{LD} \Psi^-] d\mu_L \quad (\text{C.15})$$

This integral can be evaluated analytically (by performing the μ_L integration of the piecewise H-function), yielding the known closed form:

$$\Gamma_{LD,U}(\mu' \rightarrow \mu) = \frac{\omega_{LD}}{3\pi} (\sin\beta - \beta\cos\beta) + \frac{\tau_{LD}}{3} \cos\beta \quad (\text{C.16})$$

where β is the scattering angle, $\cos\beta = \mu\mu' + \sqrt{1-\mu^2}\sqrt{1-\mu'^2}$, and $\omega_{LD} = \rho_{LD} + \tau_{LD}$.

This provides a transparent derivation of the uniform closed form (previously Eq. (C.6)), showing it as the analytical evaluation of the single-integral formula.

§15. Application to Non-Uniform Leaf Inclination Distributions

For the four smooth non-uniform distributions — planophile, erectophile, plagiophile, and extremophile — the $\bar{g}_L$ weighting prevents closed-form evaluation of the μ_L integral. However, Eq. (C.14) provides an efficient computational formula: a single Gauss-Legendre quadrature over μ_L , with the H-function evaluated analytically at each quadrature point.

Specifically, for N-point quadrature with nodes x_i and weights w_i on $[0, 1]$:

$$\Gamma_{LD}(\mu' \rightarrow \mu) \approx \sum_{i=1}^N w_i \bar{g}_L(x_i) [\tau_{LD} \Psi^+(\mu, \mu', x_i) + \rho_{LD} \Psi^-(\mu, \mu', x_i)] \quad (\text{C.17})$$

This is a single sum of N terms (typically $N = 12$ suffices), compared to the double or triple numerical integration required without the H-function.

§16. Scope and Limitations

The H-function formulation applies to uniform h_L only. When the azimuthal leaf-normal distribution is non-uniform (diaheliotropic, paraheliotropic, or general η), the azimuthal average in Eq. (C.8) does not eliminate the dependence on h_L , and the full $\Gamma_{LD}(\Omega' \rightarrow \Omega_s)$ must be computed numerically as described in the main appendix.

APPENDIX D. CONSTANT LEAF INCLINATION

The constant inclination model $\bar{u}_L = \delta(\theta_L - \theta_0)/\sin\theta_0$ concentrates all leaf normals at polar angle θ_0 , so the θ_L integration collapses by the sifting property.

§1. Problem Statement

$$\bar{g}_L(\theta_L) = \frac{\delta(\theta_L - \theta_0)}{\sin\theta_0}. \quad (\text{D.1})$$

Normalization: $\int_0^{\pi/2} [\delta(\theta_L - \theta_0)/\sin\theta_0] \sin\theta_L d\theta_L = 1$. ✓

§2. Derivation

By the sifting property of the delta function, all quantities evaluate at $\theta_L = \theta_0$. Writing $\mu_0 = \cos\theta_0$ and

$$a = \mu\mu_0, \quad b = \sin\theta\sin\theta_0, \quad a' = \mu'\mu_0, \quad b' = \sin\theta'\sin\theta_0, \quad (\text{D.2})$$

the master formula (Eq. (B.4)) gives:

$$\Gamma_{LD} = \frac{\omega_{LD}}{2} \phi_{NU} \Big|_{\theta_0} + \frac{\tau_{LD} - \rho_{LD}}{2} \Phi_{NU}^{\text{full}} \Big|_{\theta_0} \quad (\text{D.3})$$

Unrestricted integral (from Eq. (B.7)):

$$\Phi_{NU}^{\text{full}}|_{\theta_0} = \mu\mu'\mu_0^2 + \frac{1}{2}\sin\theta\sin\theta'\sin^2\theta_0\cos\delta + \frac{1}{4}\sin\theta\sin\theta'\sin^2\theta_0\cos(2\eta + \delta) \quad (\text{D.4})$$

Decomposition: $\Phi_{NU}^{\text{full}} = \Phi_U^{\text{full}} + \Delta\Phi^{\text{full}}$ with $\Phi_U^{\text{full}} = aa' + \frac{1}{2}bb'\cos\delta$ and $\Delta\Phi^{\text{full}} = \frac{1}{4}bb'\cos(2\eta + \delta)$.

Absolute-value integral:

$$\phi_{NU}|_{\theta_0} = \frac{1}{2\pi} \int_0^{2\pi} h_L |a + b\cos\alpha| |a' + b'\cos(\alpha - \delta)| d\alpha \quad (\text{D.5})$$

Branch conditions (same as G-function): $|a + b\cos\alpha|$ changes sign when $\theta + \theta_0 > \pi/2$, at $\varphi_t = \arccos(-\cot\theta\cot\theta_0)$; similarly for $|a' + b'\cos(\alpha - \delta)|$ when $\theta' + \theta_0 > \pi/2$.

Case I: $\theta + \theta_0 \leq \pi/2$ and $\theta' + \theta_0 \leq \pi/2$. Neither factor changes sign, so $\phi_{NU} = \Phi_{NU}^{\text{full}}$ and

$$\Gamma_{LD} = \tau_{LD} \Phi_{NU}^{\text{full}}|_{\theta_0}. \quad (\text{D.6})$$

Pure transmission: both beams lie on the same side of every leaf.

Case II: One factor changes sign. **Case III:** Both factors change sign (up to four transition angles).

§3. Normalization

By the universal theorem (Appendix B §5):

$$\frac{1}{\pi} \int_{4\pi} \Gamma_{LD} d\Omega = \omega_{LD} G(\Omega') \quad (\text{D.7})$$

with $G = \psi_{NU}(\mu', \mu_0, \eta)$ (G-function Appendix, Eq. (E.2)). ✓

§4. Special Cases of $\mathbf{h_L}$

Diaheliotropic ($\eta = 0$):

$$\Phi_{NU}^{\text{full}}|_{\eta=0} = \mu\mu'\mu_0^2 + \frac{3}{4}\sin\theta\sin\theta'\sin^2\theta_0\cos\delta \quad (\text{D.8})$$

Paraheliotropic ($\eta = \pi/2$):

$$\Phi_{NU}^{\text{full}}|_{\eta=\pi/2} = \mu\mu'\mu_0^2 + \frac{1}{4}\sin\theta\sin\theta'\sin^2\theta_0\cos\delta \quad (\text{D.9})$$

Dia-Para symmetry: $\Delta\Phi^{\text{full}}|_{\eta=\pi/2} = -\Delta\Phi^{\text{full}}|_{\eta=0}$. ✓

Uniform ($h_L = 1$):

$$\Phi_U^{\text{full}}|_{\theta_0} = \mu\mu'\mu_0^2 + \frac{1}{2}\sin\theta\sin\theta'\sin^2\theta_0\cos\delta \quad (\text{D.10})$$

and $\Gamma_{LD,U} = \frac{\omega_{LD}}{2}\phi_U|_{\theta_0} + \frac{\tau_{LD}-\rho_{LD}}{2}\Phi_U^{\text{full}}|_{\theta_0}$ with $G_U = \psi_U(\mu', \mu_0)$. ✓

§5. Horizontal Leaves ($\theta_0 = 0$)

With $\mu_0 = 1$, $\sin\theta_0 = 0$: $b = b' = 0$. The unrestricted integral is $\Phi_{NU}^{\text{full}} = \mu\mu'$ and the absolute-value integral is $\phi_{NU} = |\mu\mu'|$, both independent of η and δ . From the master formula:

$$\Gamma_{LD}|_{\theta_0=0} = \begin{cases} \tau_{LD}\mu\mu', & \mu\mu' > 0 \\ \rho_{LD}|\mu\mu'|, & \mu\mu' < 0 \end{cases} \quad (\text{D.11})$$

When $\mu\mu' > 0$ (same hemisphere), $|\mu\mu'| = \mu\mu'$ and the master formula gives $\frac{\omega_{LD}+\tau_{LD}-\rho_{LD}}{2}\mu\mu' = \tau_{LD}\mu\mu'$ (transmission). When $\mu\mu' < 0$ (opposite hemispheres), $|\mu\mu'| = -\mu\mu'$ and the master formula gives $\frac{-\omega_{LD}+\tau_{LD}-\rho_{LD}}{2}\mu\mu' = \rho_{LD}|\mu\mu'|$ (reflection). The azimuthal orientation is irrelevant, exactly as for $G = \mu'$ (G-function Appendix, Eq. (E.7)). ✓

§6. Vertical Leaves ($\theta_0 = \pi/2$)

With $\mu_0 = 0$, $\sin\theta_0 = 1$: $a = a' = 0$, $b = \sin\theta$, $b' = \sin\theta'$. Both factors always change sign ($\varphi_t = \varphi'_t = \pi/2$).

$$\Phi_{NU}^{\text{full}}|_{\theta_0=\pi/2} = \frac{1}{2}\sin\theta\sin\theta'\cos\delta + \frac{1}{4}\sin\theta\sin\theta'\cos(2\eta + \delta). \quad (\text{D.12})$$

Uniform h_L . The absolute-value integral factors as $\phi_U = \sin\theta\sin\theta'J(\delta)$ where

$$J(\delta) = \frac{1}{2\pi} \int_0^{2\pi} |\cos\alpha| |\cos(\alpha - \delta)| d\alpha \quad . \quad (\text{D.13})$$

Partitioning $[0, 2\pi]$ by the four sign-change points and evaluating:

$$J(\delta) = \frac{1}{2\pi} [(\pi - 2\delta)\cos\delta + 2\sin\delta], \quad 0 \leq \delta \leq \frac{\pi}{2} \quad (\text{D.14})$$

Check: $J(0) = \frac{1}{2}$, $J(\pi/2) = \frac{1}{\pi}$, $J(\pi) = \frac{1}{2}$. ✓

Substituting into the master formula and simplifying with $\omega_{LD} + \tau_{LD} - \rho_{LD} = 2\tau_{LD}$:

$$\Gamma_{LD,U}|_{\theta_0=\pi/2} = \sin\theta\sin\theta' \left[\frac{\omega_{LD}}{2\pi} (\sin\delta - \delta\cos\delta) + \frac{\tau_{LD}}{2} \cos\delta \right] \quad (\text{D.15})$$

This has the same functional form as the uniform- $\bar{g}_L$ result (Eq. (C.6)), with $\sin\theta\sin\theta'$ as prefactor, δ replacing β , and coefficients $\frac{1}{2\pi}, \frac{1}{2}$ replacing $\frac{1}{3\pi}, \frac{1}{3}$.

Normalization: $G_U = (2/\pi)\sin\theta'. \checkmark$

Diaheliotropic ($\eta = 0$): $\Phi_{NU}^{\text{full}} = \frac{3}{4}\sin\theta\sin\theta'\cos\delta$; $G = (7/3\pi)\sin\theta'. \checkmark$

Paraheliotropic ($\eta = \pi/2$): $\Phi_{NU}^{\text{full}} = \frac{1}{4}\sin\theta\sin\theta'\cos\delta$; $G = (5/3\pi)\sin\theta'. \checkmark$

Dia-Para symmetry: $\Delta\Phi^{\text{full}}|_{\eta=\pi/2} = -\Delta\Phi^{\text{full}}|_{\eta=0}. \checkmark$

§7. Summary

Master formula. $\Gamma_{LD} = \frac{\omega_{LD}}{2} \Phi_{NU} \Big|_{\theta_0} + \frac{\tau_{LD} - \rho_{LD}}{2} \Phi_{NU}^{\text{full}} \Big|_{\theta_0}$, all at $\theta_L = \theta_0$.

Unrestricted integral. $\Phi_{NU}^{\text{full}} = \mu\mu'\mu_0^2 + \frac{1}{2}bb'\cos\delta + \frac{1}{4}bb'\cos(2\eta + \delta)$.

Case I. $\theta + \theta_0 \leq \pi/2, \theta' + \theta_0 \leq \pi/2$: $\Gamma_{LD} = \tau_{LD} \Phi_{NU}^{\text{full}}$ (pure transmission).

Horizontal ($\theta_0 = 0$). $\Gamma_{LD} = \tau_{LD} \mu\mu'$ ($\mu\mu' > 0$); $\Gamma_{LD} = \rho_{LD} |\mu\mu'|$ ($\mu\mu' < 0$).

Vertical, uniform h_L ($\theta_0 = \pi/2$). $\Gamma_{LD,U} = \sin\theta\sin\theta' \left[\frac{\omega_{LD}}{2\pi} (\sin\delta - \delta\cos\delta) + \frac{\tau_{LD}}{2} \cos\delta \right]$.

Dia-Para symmetry. $\Delta\Phi^{\text{full}}|_{\eta=\pi/2} = -\Delta\Phi^{\text{full}}|_{\eta=0}$.

Universal normalization. $\frac{1}{\pi} \int_{4\pi} \Gamma_{LD} d\Omega = \omega_{LD} G(\mathbf{\Omega}')$ with $G = \psi_{NU}(\mu', \mu_0, \eta)$.

APPENDIX E. SMOOTH NON-UNIFORM LEAF INCLINATION DISTRIBUTIONS

For each smooth non-uniform inclination distribution, the derivation follows the same pattern: compute the moment M2, integrate the master formula against $\bar{u}_L$, and verify normalization via the universal theorem.

§1. Planophile Leaf Inclination

$$\bar{g}_L(\theta_L) = 3\mu_L^2. \quad (\text{E.1})$$

Normalization: $3 \int_0^1 u^2 du = 1. \checkmark$

Derivation:

Integrating Eq. (B.7) with $\bar{g}_L = 3\mu_L^2$, the required moments are:

$$M_2 = \int_0^{\pi/2} \bar{g}_L \mu_L^2 \sin\theta_L d\theta_L = \frac{3}{5}, \quad M_s = 1 - M_2 = \frac{2}{5} \quad (\text{E.2})$$

The unrestricted integral after θ_L integration becomes:

$$\Gamma^{\text{full}} = \frac{3}{5} \mu\mu' + \frac{1}{5} \sin\theta \sin\theta' \cos\delta + \frac{1}{10} \sin\theta \sin\theta' \cos(2\eta + \delta) \quad (\text{E.3})$$

Decomposing $\Gamma^{\text{full}} = \Gamma_U^{\text{full}} + \Delta\Gamma^{\text{full}}$:

$$\Gamma_U^{\text{full}} = \frac{3}{5} \cos\beta - \frac{2}{5} \sin\theta \sin\theta' \cos\delta \quad (\text{E.4})$$

$$\Delta\Gamma^{\text{full}} = \frac{1}{10} \sin\theta \sin\theta' \cos(2\eta + \delta), \quad (\text{E.5})$$

The master formula (Eq. (B.4)) gives:

$$\Gamma_{LD} = \frac{\omega_{LD}}{2} \Phi^{|\cdot|} + \frac{\tau_{LD} - \rho_{LD}}{2} \Gamma^{\text{full}} \quad (\text{E.6})$$

where the absolute-value integral after θ_L integration is:

$$\Phi^{|\cdot|} = \int_0^{\pi/2} \bar{g}_L \phi_{NU} \sin\theta_L d\theta_L \quad (\text{E.7})$$

with ϕ_{NU} from Eq. (C.10). Unlike uniform $\bar{g}_L$ (where the axis-alignment trick yields a closed form in β), the weighting by $3\mu_L^2$ prevents closed-form evaluation; $\Phi^{|\cdot|}$ must be evaluated numerically or left as an integral, exactly as ΔG in the G-function Appendix.

Normalization:

By the universal theorem (Appendix B §5):

$$\frac{1}{\pi} \int_{4\pi} \Gamma_{LD} d\Omega = \omega_{LD} G(\Omega') \quad (\text{E.8})$$

with $G = \frac{3(1+\mu'^2)}{8} + \Delta G$ from the G-function Appendix (Eq. (E.12)). ✓

Special Cases of $\mathbf{h}_L$

Diaheliotropic ($\eta = 0$):

$$\Gamma^{\text{full}}|_{\eta=0} = \frac{3}{5} \mu\mu' + \frac{3}{10} \sin\theta \sin\theta' \cos\delta \quad (\text{E.9})$$

Normalization: $G = \frac{3(1+\mu'^2)}{8} + \Delta G|_{\eta=0}$. ✓

Paraheliotropic ($\eta = \pi/2$):

$$\Gamma^{\text{full}}|_{\eta=\pi/2} = \frac{3}{5} \mu\mu' + \frac{1}{10} \sin\theta \sin\theta' \cos\delta \quad (\text{E.10})$$

Dia-Para symmetry: $\Delta\Gamma^{\text{full}}|_{\eta=\pi/2} = -\Delta\Gamma^{\text{full}}|_{\eta=0}$. ✓

Uniform ($h_L = 1$):

$$\Gamma_U^{\text{full}} = \frac{3}{5} \mu \mu' + \frac{1}{5} \sin \theta \sin \theta' \cos \delta \quad (\text{E.11})$$

$$\Gamma_{LD,U} = \frac{\omega_{LD}}{2} \Phi_U^{|\cdot|} + \frac{\tau_{LD} - \rho_{LD}}{2} \Gamma_U^{\text{full}} \text{ with } G_U = \frac{3(1+\mu'^2)}{8}. \checkmark$$

Summary:

Moments. $M_2 = \frac{3}{5}$, $M_s = \frac{2}{5}$.

Unrestricted integral. $\Gamma^{\text{full}} = \frac{3}{5} \mu \mu' + \frac{1}{5} \sin \theta \sin \theta' \cos \delta + \frac{1}{10} \sin \theta \sin \theta' \cos(2\eta + \delta)$.

Uniform decomposition. $\Gamma_U^{\text{full}} = \frac{3}{5} \cos \beta - \frac{2}{5} \sin \theta \sin \theta' \cos \delta$.

Dia-Para symmetry. $\Delta \Gamma^{\text{full}}|_{\eta=\pi/2} = -\Delta \Gamma^{\text{full}}|_{\eta=0}$.

Universal normalization. $\frac{1}{\pi} \int_{4\pi} \Gamma_{LD} d\Omega = \omega_{LD} G(\mathbf{\Omega}')$.

§2. Erectophile Leaf Inclination Γ_{LD}

Problem Statement:

$$\bar{g}_L(\theta_L) = \frac{3}{2} \sin^2 \theta_L. \quad (\text{E.12})$$

Normalization: $\frac{3}{2} \int_0^1 (1 - u^2) du = 1. \checkmark$

Derivation:

Integrating Eq. (B.7) with $\bar{g}_L = \frac{3}{2} \sin^2 \theta_L$, the required moments are:

$$M_2 = \int_0^{\pi/2} \bar{g}_L \mu_L^2 \sin \theta_L d\theta_L = \frac{1}{5}, \quad M_s = 1 - M_2 = \frac{4}{5} \quad (\text{E.13})$$

The unrestricted integral after θ_L integration becomes:

$$\Gamma^{\text{full}} = \frac{1}{5} \mu \mu' + \frac{2}{5} \sin \theta \sin \theta' \cos \delta + \frac{1}{5} \sin \theta \sin \theta' \cos(2\eta + \delta) \quad (\text{E.14})$$

Decomposing $\Gamma^{\text{full}} = \Gamma_U^{\text{full}} + \Delta \Gamma^{\text{full}}$:

$$\Gamma_U^{\text{full}} = \frac{1}{5} \cos \beta + \frac{1}{5} \sin \theta \sin \theta' \cos \delta \quad (\text{E.15})$$

$$\Delta \Gamma^{\text{full}} = \frac{1}{5} \sin \theta \sin \theta' \cos(2\eta + \delta), \quad (\text{E.16})$$

The master formula (Eq. (B.4)) gives:

$$\Gamma_{LD} = \frac{\omega_{LD}}{2} \Phi^{|\cdot|} + \frac{\tau_{LD} - \rho_{LD}}{2} \Gamma^{\text{full}} \quad (\text{E.17})$$

where the absolute-value integral after θ_L integration is:

$$\Phi^{|\cdot|} = \int_0^{\pi/2} \bar{g}_L \phi_{NU} \sin\theta_L d\theta_L \quad (\text{E.18})$$

with ϕ_{NU} from Eq. (C.10). Unlike uniform $\bar{g}_L$ (where the axis-alignment trick yields a closed form in β), the weighting by $\frac{3}{2}\sin^2\theta_L$ prevents closed-form evaluation; $\Phi^{|\cdot|}$ must be evaluated numerically or left as an integral, exactly as ΔG in the G-function Appendix.

Normalization:

By the universal theorem (Appendix B §5):

$$\frac{1}{\pi} \int_{4\pi} \Gamma_{LD} d\Omega = \omega_{LD} G(\mathbf{\Omega}') \quad (\text{E.19})$$

with $G = \frac{3(2+\sin^2\theta')}{16} + \Delta G$ from the G-function Appendix (Eq. (E.15)). ✓

Special Cases of $\mathbf{h}_L$

Diaheliotropic ($\eta = 0$):

$$\Gamma^{\text{full}}|_{\eta=0} = \frac{1}{5} \mu\mu' + \frac{3}{5} \sin\theta\sin\theta'\cos\delta \quad (\text{E.20})$$

Normalization: $G = \frac{3(2+\sin^2\theta')}{16} + \Delta G|_{\eta=0}$. ✓

Paraheliotropic ($\eta = \pi/2$):

$$\Gamma^{\text{full}}|_{\eta=\pi/2} = \frac{1}{5} \mu\mu' + \frac{1}{5} \sin\theta\sin\theta'\cos\delta \quad (\text{E.21})$$

Dia-Para symmetry: $\Delta\Gamma^{\text{full}}|_{\eta=\pi/2} = -\Delta\Gamma^{\text{full}}|_{\eta=0}$. ✓

Uniform ($h_L = 1$):

$$\Gamma_U^{\text{full}} = \frac{1}{5} \mu\mu' + \frac{2}{5} \sin\theta\sin\theta'\cos\delta \quad (\text{E.22})$$

$$\Gamma_{LD,U} = \frac{\omega_{LD}}{2} \Phi^{|\cdot|} + \frac{\tau_{LD}-\rho_{LD}}{2} \Gamma_U^{\text{full}} \text{ with } G_U = \frac{3(2+\sin^2\theta')}{16}. \checkmark$$

Summary:

Moments. $M_2 = \frac{1}{5}$, $M_s = \frac{4}{5}$.

Unrestricted integral. $\Gamma^{\text{full}} = \frac{1}{5} \mu\mu' + \frac{2}{5} \sin\theta\sin\theta'\cos\delta + \frac{1}{5} \sin\theta\sin\theta'\cos(2\eta + \delta)$.

Uniform decomposition. $\Gamma_U^{\text{full}} = \frac{1}{5} \cos\beta + \frac{1}{5} \sin\theta\sin\theta'\cos\delta$.

Dia-Para symmetry. $\Delta\Gamma^{\text{full}}|_{\eta=\pi/2} = -\Delta\Gamma^{\text{full}}|_{\eta=0}$.

Universal normalization. $\frac{1}{\pi} \int_{4\pi} \Gamma_{LD} d\Omega = \omega_{LD} G(\mathbf{\Omega}')$.

§3. Plagiophile Leaf Inclination Γ_{LD}

Problem Statement:

$$\bar{g}_L(\theta_L) = \frac{15}{8} \sin^2 2\theta_L. \quad (\text{E.23})$$

Normalization: $\frac{15}{2} \int_0^1 u^2 (1 - u^2) du = 1. \checkmark$

Derivation:

Integrating Eq. (B.7) with $\bar{g}_L = \frac{15}{8} \sin^2 2\theta_L$, the required moments are:

$$M_2 = \int_0^{\pi/2} \bar{g}_L \mu_L^2 \sin\theta_L d\theta_L = \frac{3}{7}, \quad M_s = 1 - M_2 = \frac{4}{7} \quad (\text{E.24})$$

The unrestricted integral after θ_L integration becomes:

$$\Gamma^{\text{full}} = \frac{3}{7} \mu\mu' + \frac{2}{7} \sin\theta \sin\theta' \cos\delta + \frac{1}{7} \sin\theta \sin\theta' \cos(2\eta + \delta) \quad (\text{E.25})$$

Decomposing $\Gamma^{\text{full}} = \Gamma_U^{\text{full}} + \Delta\Gamma^{\text{full}}$:

$$\Gamma_U^{\text{full}} = \frac{3}{7} \cos\beta - \frac{1}{7} \sin\theta \sin\theta' \cos\delta \quad (\text{E.26})$$

$$\Delta\Gamma^{\text{full}} = \frac{1}{7} \sin\theta \sin\theta' \cos(2\eta + \delta), \quad (\text{E.27})$$

The master formula (Eq. (B.4)) gives:

$$\Gamma_{LD} = \frac{\omega_{LD}}{2} \Phi^{|\cdot|} + \frac{\tau_{LD} - \rho_{LD}}{2} \Gamma^{\text{full}} \quad (\text{E.28})$$

where the absolute-value integral after θ_L integration is:

$$\Phi^{|\cdot|} = \int_0^{\pi/2} \bar{g}_L \phi_{NU} \sin\theta_L d\theta_L \quad (\text{E.29})$$

with ϕ_{NU} from Eq. (C.10). Unlike uniform $\bar{g}_L$ (where the axis-alignment trick yields a closed form in β), the weighting by $\frac{15}{8} \sin^2 2\theta_L$ prevents closed-form evaluation; $\Phi^{|\cdot|}$ must be evaluated numerically or left as an integral, exactly as ΔG in the G-function Appendix.

Normalization:

By the universal theorem (Appendix B §5):

$$\frac{1}{\pi} \int_{4\pi} \Gamma_{LD} d\Omega = \omega_{LD} G(\Omega') \quad (\text{E.30})$$

with $G = \frac{5(3+\mu'^4)}{32} + \Delta G$ from the G-function Appendix (Eq. (E.17)). $\checkmark$

Special Cases of $\mathbf{h}_L$

Diaheliotropic ($\eta = 0$):

$$\Gamma^{\text{full}}|_{\eta=0} = \frac{3}{7} \mu\mu' + \frac{3}{7} \sin\theta \sin\theta' \cos\delta \quad (\text{E.31})$$

Normalization: $G = \frac{5(3+\mu'^4)}{32} + \Delta G|_{\eta=0} \cdot \checkmark$

Paraheliotropic ($\eta = \pi/2$):

$$\Gamma^{\text{full}}|_{\eta=\pi/2} = \frac{3}{7} \mu\mu' + \frac{1}{7} \sin\theta \sin\theta' \cos\delta \quad (\text{E.32})$$

Dia-Para symmetry: $\Delta\Gamma^{\text{full}}|_{\eta=\pi/2} = -\Delta\Gamma^{\text{full}}|_{\eta=0} \cdot \checkmark$

Uniform ($h_L = 1$):

$$\Gamma_U^{\text{full}} = \frac{3}{7} \mu\mu' + \frac{2}{7} \sin\theta \sin\theta' \cos\delta \quad (\text{E.33})$$

$$\Gamma_{LD,U} = \frac{\omega_{LD}}{2} \Phi_U^{| \cdot |} + \frac{\tau_{LD} - \rho_{LD}}{2} \Gamma_U^{\text{full}} \text{ with } G_U = \frac{5(3+\mu'^4)}{32} \cdot \checkmark$$

Summary:

Moments. $M_2 = \frac{3}{7}, M_s = \frac{4}{7}$.

Unrestricted integral. $\Gamma^{\text{full}} = \frac{3}{7} \mu\mu' + \frac{2}{7} \sin\theta \sin\theta' \cos\delta + \frac{1}{7} \sin\theta \sin\theta' \cos(2\eta + \delta)$.

Uniform decomposition. $\Gamma_U^{\text{full}} = \frac{3}{7} \cos\beta - \frac{1}{7} \sin\theta \sin\theta' \cos\delta$.

Dia-Para symmetry. $\Delta\Gamma^{\text{full}}|_{\eta=\pi/2} = -\Delta\Gamma^{\text{full}}|_{\eta=0}$.

Universal normalization. $\frac{1}{\pi} \int_{4\pi} \Gamma_{LD} d\Omega = \omega_{LD} G(\Omega')$.

§4. Extremophile Leaf Inclination $\mathbf{F}_{LD}$

Problem Statement:

$$\bar{g}_L(\theta_L) = \frac{15}{7} \cos^2 2\theta_L. \quad (\text{E.34})$$

Normalization: $\frac{15}{7} \int_0^1 (2u^2 - 1)^2 du = 1 \cdot \checkmark$

Derivation:

Integrating Eq. (B.7) with $\bar{g}_L = \frac{15}{7} \cos^2 2\theta_L$, the required moments are:

$$M_2 = \int_0^{\pi/2} \bar{g}_L \mu_L^2 \sin\theta_L d\theta_L = \frac{11}{49}, \quad M_s = 1 - M_2 = \frac{38}{49} \quad (\text{E.35})$$

The unrestricted integral after θ_L integration becomes:

$$\Gamma^{\text{full}} = \frac{11}{49} \mu\mu' + \frac{19}{49} \sin\theta \sin\theta' \cos\delta + \frac{19}{98} \sin\theta \sin\theta' \cos(2\eta + \delta) \quad (\text{E.36})$$

Decomposing $\Gamma^{\text{full}} = \Gamma_U^{\text{full}} + \Delta\Gamma^{\text{full}}$:

$$\Gamma_U^{\text{full}} = \frac{11}{49} \cos\beta + \frac{8}{49} \sin\theta \sin\theta' \cos\delta \quad (\text{E.37})$$

$$\Delta\Gamma^{\text{full}} = \frac{19}{98} \sin\theta \sin\theta' \cos(2\eta + \delta), \quad (\text{E.38})$$

The master formula (Eq. (B.4)) gives:

$$\Gamma_{LD} = \frac{\omega_{LD}}{2} \Phi^{|\cdot|} + \frac{\tau_{LD} - \rho_{LD}}{2} \Gamma^{\text{full}} \quad (\text{E.39})$$

where the absolute-value integral after θ_L integration is:

$$\Phi^{|\cdot|} = \int_0^{\pi/2} \bar{g}_L \phi_{NU} \sin\theta_L d\theta_L \quad (\text{E.40})$$

with ϕ_{NU} from Eq. (C.10). Unlike uniform $\bar{g}_L$ (where the axis-alignment trick yields a closed form in β), the weighting by $\frac{15}{7} \cos^2 2\theta_L$ prevents closed-form evaluation; $\Phi^{|\cdot|}$ must be evaluated numerically or left as an integral, exactly as ΔG in the G-function Appendix.

Normalization:

By the universal theorem (Appendix B §5):

$$\frac{1}{\pi} \int_{4\pi} \Gamma_{LD} d\Omega = \omega_{LD} G(\Omega') \quad (\text{E.41})$$

with $G = \frac{5(3-\mu'^4)}{28} + \Delta G$ from the G-function Appendix (Eq. (E.19)). ✓

Special Cases of $\mathbf{h}_L$

Diaheliotropic ($\eta = 0$):

$$\Gamma^{\text{full}}|_{\eta=0} = \frac{11}{49} \mu\mu' + \frac{57}{98} \sin\theta \sin\theta' \cos\delta \quad (\text{E.42})$$

Normalization: $G = \frac{5(3-\mu'^4)}{28} + \Delta G|_{\eta=0}$. ✓

Paraheliotropic ($\eta = \pi/2$):

$$\Gamma^{\text{full}}|_{\eta=\pi/2} = \frac{11}{49} \mu\mu' + \frac{19}{98} \sin\theta \sin\theta' \cos\delta \quad (\text{E.43})$$

Dia-Para symmetry: $\Delta\Gamma^{\text{full}}|_{\eta=\pi/2} = -\Delta\Gamma^{\text{full}}|_{\eta=0}$. ✓

Uniform ($h_L = 1$):

$$\Gamma_U^{\text{full}} = \frac{11}{49} \mu\mu' + \frac{19}{49} \sin\theta \sin\theta' \cos\delta \quad (\text{E.44})$$

$$\Gamma_{LD,U} = \frac{\omega_{LD}}{2} \Phi_U^{|\cdot|} + \frac{\tau_{LD} - \rho_{LD}}{2} \Gamma_U^{\text{full}} \text{ with } G_U = \frac{5(3-\mu'^4)}{28}. \checkmark$$

Summary:

Moments. $M_2 = \frac{11}{49}, M_s = \frac{38}{49}.$

Unrestricted integral. $\Gamma^{\text{full}} = \frac{11}{49}\mu\mu' + \frac{19}{49}\sin\theta\sin\theta'\cos\delta + \frac{19}{98}\sin\theta\sin\theta'\cos(2\eta + \delta).$

Uniform decomposition. $\Gamma_U^{\text{full}} = \frac{11}{49}\cos\beta + \frac{8}{49}\sin\theta\sin\theta'\cos\delta.$

Dia–Para symmetry. $\Delta\Gamma^{\text{full}}|_{\eta=\pi/2} = -\Delta\Gamma^{\text{full}}|_{\eta=0}.$

Universal normalization. $\frac{1}{\pi}\int_{4\pi}\Gamma_{LD} d\Omega = \omega_{LD} G(\mathbf{\Omega}').$

APPENDIX F. SUMMARY OF ALL RESULTS

This appendix collects in one place all analytical results for the area scattering phase function Γ_{LD} derived throughout the preceding appendices.

§1. Moment Structure

For any leaf inclination distribution $\bar{g}_L$, the unrestricted integral after θ_L integration takes the universal form

$$\Gamma^{\text{full}} = M_2 \mu\mu' + \frac{M_s}{2} \sin\theta\sin\theta'\cos\delta + \frac{M_s}{4} \sin\theta\sin\theta'\cos(2\eta + \delta), \quad (\text{F.1})$$

where $M_2 = \int_0^{\pi/2} \bar{g}_L \mu_L^2 \sin\theta_L d\theta_L$ and $M_s = 1 - M_2$.

Distribution	$\bar{g}_L$	M_2	M_s
Uniform	1	1/3	2/3
Constant	$\delta(\theta_L - \theta_0)/\sin\theta_0$	μ_0^2	$\sin^2\theta_0$
Planophile	$3\mu_L^2$	3/5	2/5
Erectophile	$\frac{3}{2}\sin^2\theta_L$	1/5	4/5
Plagiophile	$\frac{15}{8}\sin^2 2\theta_L$	3/7	4/7
Extremophile	$\frac{15}{7}\cos^2 2\theta_L$	11/49	38/49

§2. Structural Properties

Master formula. $\Gamma_{LD} = \frac{\omega_{LD}}{2} \Phi^{|\cdot|} + \frac{\tau_{LD} - \rho_{LD}}{2} \Gamma^{\text{full}},$ valid for all $\bar{g}_L$ and h_L .

Dia–Para symmetry. $\Delta\Gamma^{\text{full}}|_{\eta=\pi/2} = -\Delta\Gamma^{\text{full}}|_{\eta=0}$ for every $\bar{g}_L$.

Universal normalization. $\frac{1}{\pi}\int_{4\pi}\Gamma_{LD} d\Omega = \omega_{LD} G(\mathbf{\Omega}')$ for every $\bar{g}_L$ and h_L .

Closed forms. $\Phi^{|\cdot|}$ admits closed-form evaluation only for uniform $\bar{g}_L$ (Eq. (C.5)) and constant $\bar{g}_L$ with $\theta_0 = \pi/2$ (Eq. (D.15)). For all other distributions, $\Phi^{|\cdot|}$ must be evaluated numerically or left as an integral. For uniform h_L , the H-function formulation (Appendix C) reduces the computation to a single quadrature over μ_L with an analytical kernel.

Scattering-angle dependence. For uniform $\bar{g}_L$ with uniform h_L , $\Gamma_{LD,U}$ depends only on the scattering angle β .